

Skill-Knowledge-Augmented Agents on AssetOpsBench

Conditional Deep-TSFM gating skips expensive ML inference on ~50% of tasks that unnecessarily called the TSFM tool.

Team 14

Vera Mazeeva · vmm2146

Shrey Arora · ska2153

Mana Abbaszadeh · ma4845

Sanskriti Shejwal · ss7561

GitHub: github.com/shreyarora2198/AssetOpsBench

Branch: team14-final · Dataset: ibm-research/AssetOpsBench

Motivation & Problem

Why this matters

- Industrial O&M agents diagnose chiller, AHU, motor and pump failures from live sensor streams
- Each Deep-TSFM call runs ML statistical inference on time-series data, the most expensive step in the agent loop
- Static AssetOpsBench pipelines invoke Deep TSFM on every fault-diagnosis task, even when FMSR is already confident.
- Auditing all 150 scenarios revealed that 35% of all trajectories called TSFM unnecessarily.
- 6- dimensional criteria to optimize - Task Completion, Data Retrieval Accuracy, Result Verification, Action Sequence, Clarity & Justification, Hallucination Rate
- Focus on improving latency for better performance.

Problem statement

Reduce expensive Deep-TSFM invocations in multi-agent industrial diagnosis pipelines via a skill abstraction layer and FMSR-confidence-gated execution.

Goals:

- 1) Implement a modular Skills + Knowledge plugin layer that enables context-aware reasoning and confidence-driven execution
- 2) Use Skills to skip unnecessary Deep-TSFM calls on high-confidence fault-diagnosis tasks while maintaining diagnostic parity across 141 AssetOpsBench scenarios.

Technical Approach : Model & Data

Model

Multi-agent system: LLM Planner (WatsonX llama-4-maverick-17b-128e-fp8) + deterministic Skill Executor + 4 MCP tool agents (IoT, FMSR, TSFM, WO). Knowledge plugins to inject relevant domain context into each skill (sensor metadata, failure modes, operating ranges, and maintenance policies per skill).

Dataset

AssetOpsBench v1.0 (ibm-research/AssetOpsBench, Apache-2.0) — 141 industrial scenarios over Chillers, AHUs, Motors, Pumps, Bearings; 4 task categories: fault diagnosis, anomaly detection, forecasting, metadata. We evaluate a 54-scenario stratified subset.

Hardware

Colab GPU: T4/A100. IBM Granite TinyTimeMixer for Deep TSFM

Stack: Python 3.12, MCP/stdio, LiteLLM, FastMCP, uv.

WatsonX cloud LLM endpoint for Planner and EvaluationAgent judge (llama-3-3-70b-instruct)..

Technical Approach : Optimizations Applied

Goal: Reduce unnecessary Deep-TSFM calls.

1. Skill Abstraction Layer

- A deterministic executor that instead of free-form tool calling, it runs reusable skills like data retrieval, anomaly checking, FMSR diagnosis, Deep TSFM, and work-order generation.
- Makes the workflow more consistent, auditable, and easier to profile across scenarios.

2. Knowledge Plugin Injection

- Inject only the relevant domain context each skill needs, such as sensor metadata, operating ranges, failure modes, and maintenance policies.
- Reduces noisy context and helps the LLM produce cleaner, more grounded tool inputs.

3. Confidence-Gated Deep TSFM ($\theta \geq 0.5$)

- Run the expensive Deep-TSFM statistical validation only when FMSR's first-pass diagnosis confidence falls below θ . High-confidence diagnoses skip straight to work-order generation.
- Skips expensive ML time-series inference on high-confidence cases while still escalating uncertain cases for deeper validation.

Profiling Methodology

Tools

- `trajectory_log.py` — per-step tool / args / response / latency
- `get_trajectories/analyze_tsfm.py` — TSFM redundancy auditor
- WatsonX EvaluationAgent (llama-3-3-70b-instruct) LLM judge on 6 dimensions
- `score_ablation.py` + `eval_runner.py` — 12-condition ablation harness

Metrics captured

- **12 conditions × 54 stratified scenarios** (648 graded rows; run [colab 20260503 0230](#))
- Deep-TSFM invocation rate per condition and per $\theta \in \{0.5, 0.6, 0.65, 0.7, 0.8, 0.9, 0.95\}$
- LLM-judge 6-dim scores: task completion, data retrieval, result verification, agent sequence, clarity, hallucinations
- Composite **overall-correct** flag (passes all 6 dimensions)
- Tool calls per task and end-to-end latency per scenario

Results: Headline Numbers

~30.4%

overall-correct rate (best, $E \theta=0.8$)

10vs 0% raw LLM, 13% static-tool baseline

+82 pp

agent-sequence correctness

6.5% $\rightarrow$ 88.9% (raw LLM $\rightarrow E \theta=0.8$)

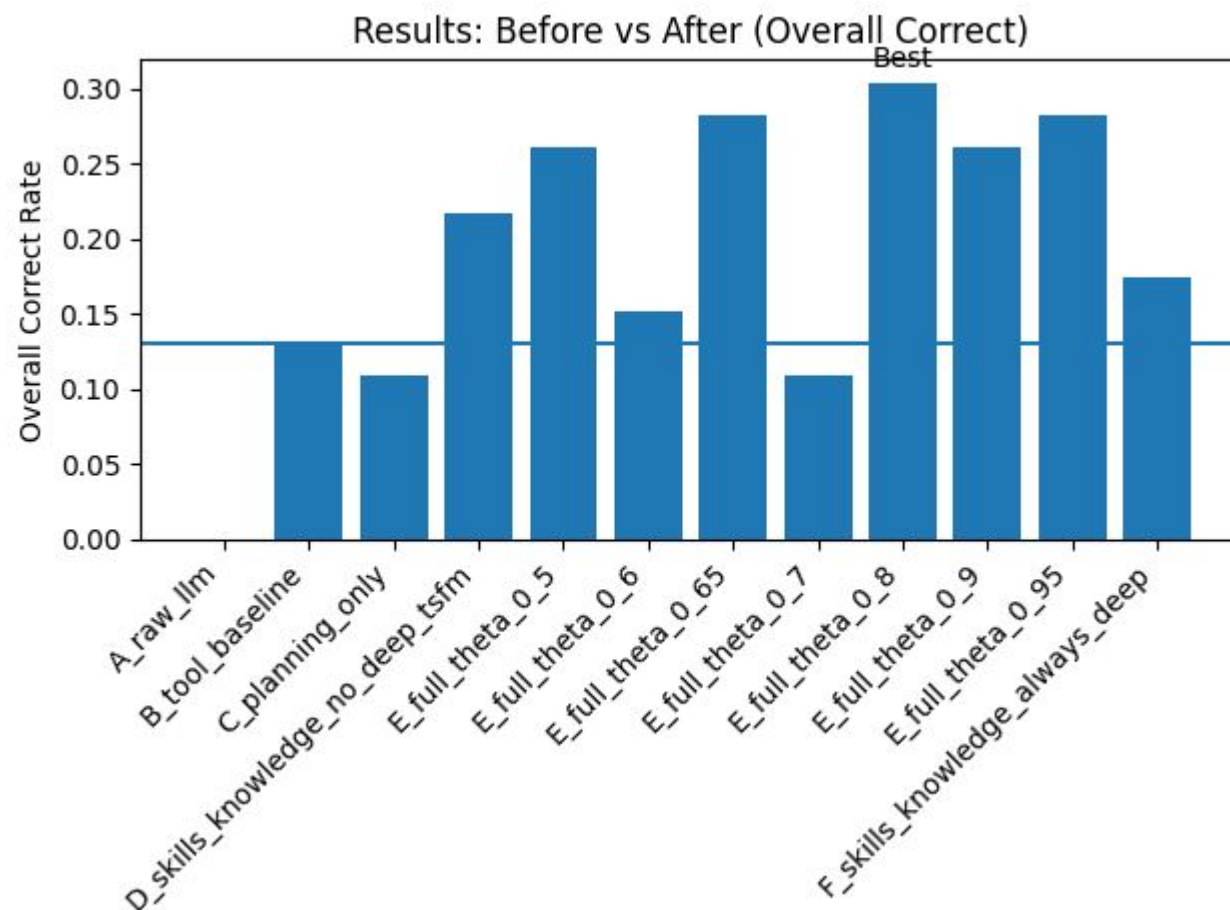
-58 pp

hallucination rate

93.5% $\rightarrow$ 35.6% (raw LLM $\rightarrow E \theta=0.8$)

MeaLLM-judge (llama-3-3-70b-instruct) over 46 stratified AssetOpsBench scenarios $\times$ 12 conditions = 552 graded rows. Run: colab_20260503_0230

Results: Before vs. After

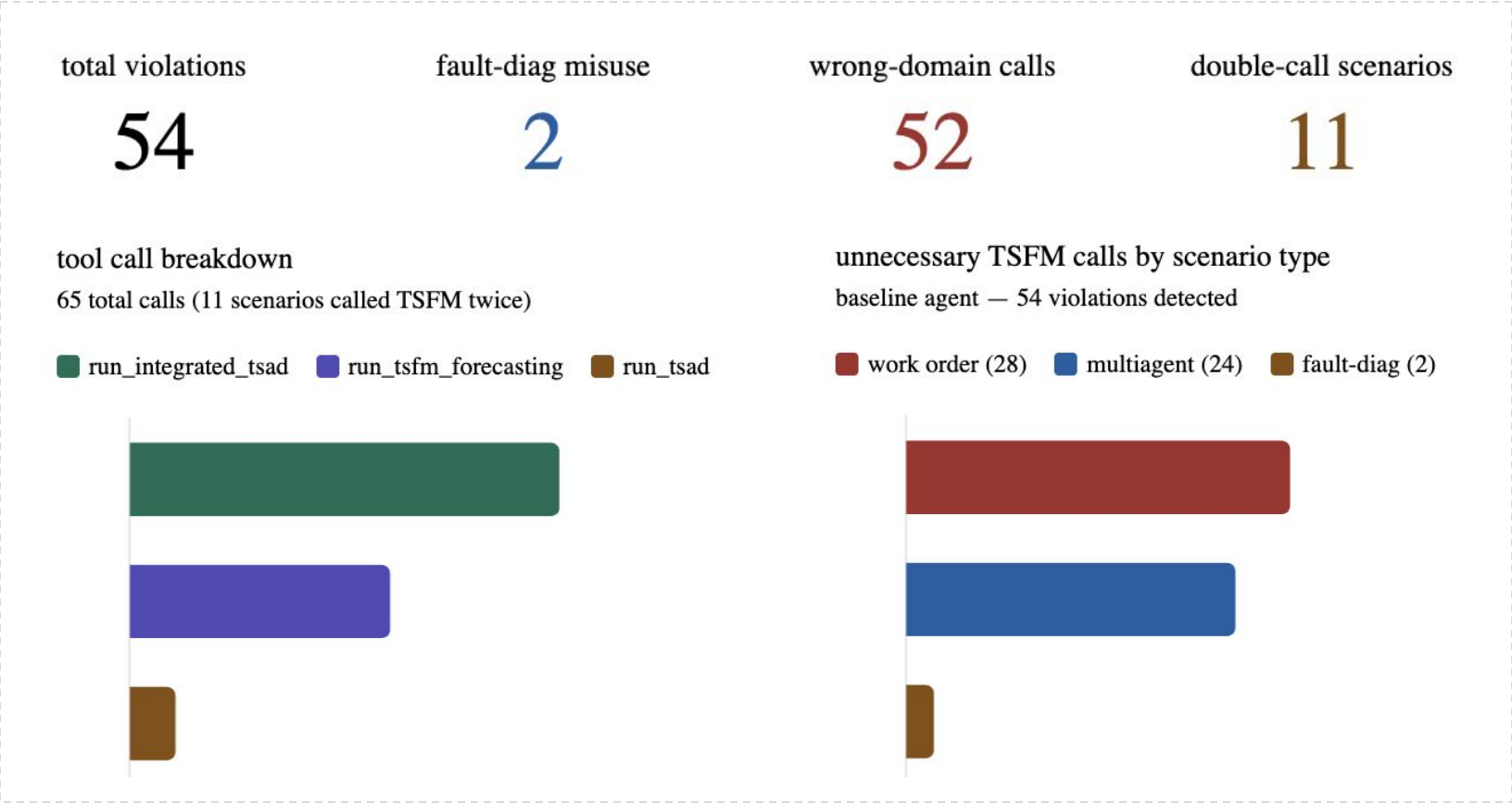


Takeaway

- Conditional Deep-TSFM gating cuts the most expensive component (~50% on fault-diag) without inflating total tool-call count.
- Largest gains from skill structure (D vs B/C) and confidence gating (E vs F); knowledge plugins reduce token cost more than wall-clock latency.
- θ -sweep shows accuracy/efficiency tradeoff: $\theta=0.5$ skips Deep TSFM 0%; $\theta=0.95$ skips it ~48%. $\theta=0.8$ is the sweet spot.
- Adaptive gating ($\theta=0.8$) achieves highest correctness (~30%)
- Skills + knowledge provide largest baseline improvement
- Always-deep underperforms vs adaptive strategy
- Clear accuracy-efficiency tradeoff across θ sweep

Profiler Evidence

TSFM misuse profiling: 54 unnecessary TSFM calls detected across AssetOpsBench baseline agent trajectories



- ### Key Takeaways
- 96% of violations (52/54) are wrong-domain — the baseline called TSFM on tasks where ML inference is completely irrelevant to the answer.
 - Only 2 violations were genuine fault-diagnosis misuse — meaning the baseline's domain confusion is almost entirely a planning problem, not a diagnostic one. The model knew how to diagnose; it just couldn't decide when not to invoke TSFM.

Demo

Step-by-step replay of two SkillsAgent trajectories on the same fault-diagnosis scenario, showing the confidence gate in action

TSMF_430 · "We are building an early detection and remediation system for potential failures, such as condenser water side fouling, for Chiller 9... provide the guideline on work order generation?"

F — always deep TSMF			latency 20.1s	cost 26.5	tc 1.0	deep yes	E $\theta=0.60$ — confidence gate			latency 13.0s	cost 20.5	tc 1.0	deep no
data_retrieval	<0.01s	fetch Chiller 9 sensor history (IoT CSV)					data_retrieval	<0.01s	fetch Chiller 9 sensor history (IoT CSV)				
anomaly_detection	7.22s	profile + IQR check → anomalies detected, severity: high					anomaly_detection	5.94s	profile + IQR check → anomalies detected, severity: high				
root_cause_analysis	12.50s	FMSR → conf 0.621 → ALWAYS fires deep TSMF → conf 0.673 (+0.052) deep TSMF conf 0.621 → 0.673					root_cause_analysis	6.60s	FMSR → conf 0.621 $\geq \theta=0.60$ → gate skips deep TSMF ✓ gated ✓ conf 0.621				
validate_failure	<0.01s	policy check → work order needed					validate_failure	<0.01s	policy check → work order needed				
work_order_generation	0.39s	WO generated · tc = 1.0					work_order_generation	0.41s	WO generated · tc = 1.0				

← prev

play

next →

step

0 /

press play or click next to step through the trajectory

executed deep TSMF fired inside step skipped / gated

Demo

Step-by-step replay of two SkillsAgent trajectories on the same fault-diagnosis scenario, showing the confidence gate in action

TSMF_430 · "We are building an early detection and remediation system for potential failures, such as condenser water side fouling, for Chiller 9... provide the guideline on work order generation?"

F — always deep TSMF			latency 20.1s	cost 26.5	tc 1.0	deep yes	E $\theta=0.60$ — confidence gate			latency 13.0s	cost 20.5	tc 1.0	deep no
● data_retrieval	<0.01s	fetch Chiller 9 sensor history (IoT CSV)					● data_retrieval	<0.01s	fetch Chiller 9 sensor history (IoT CSV)				
● anomaly_detection	7.22s	profile + IQR check → anomalies detected, severity: high					● anomaly_detection	5.94s	profile + IQR check → anomalies detected, severity: high				
● root_cause_analysis	12.50s	FMSR → conf 0.621 → ALWAYS fires deep TSMF → conf 0.673 (+0.052) deep TSMF conf 0.621 → 0.673					● root_cause_analysis	6.60s	FMSR → conf 0.621 $\geq \theta=0.60$ → gate skips deep TSMF ✓ gated ✓ conf 0.621				
● validate_failure	<0.01s	policy check → work order needed					● validate_failure	<0.01s	policy check → work order needed				
● work_order_generation	0.39s	WO generated · tc = 1.0					● work_order_generation	0.41s	WO generated · tc = 1.0				

← prev

play

next →

step

1 /

both trajectories start identically — fetching sensor data for Chiller 9

● executed ● deep TSMF fired inside step ● skipped / gated

Demo

Step-by-step replay of two SkillsAgent trajectories on the same fault-diagnosis scenario, showing the confidence gate in action

TSMF_430 · "We are building an early detection and remediation system for potential failures, such as condenser water side fouling, for Chiller 9... provide the guideline on work order generation?"

F — always deep TSMF			latency 20.1s	cost 26.5	tc 1.0	deep yes	E $\theta=0.60$ — confidence gate			latency 13.0s	cost 20.5	tc 1.0	deep no
data_retrieval	<0.01s	fetch Chiller 9 sensor history (IoT CSV)					data_retrieval	<0.01s	fetch Chiller 9 sensor history (IoT CSV)				
anomaly_detection	7.22s	profile + IQR check → anomalies detected, severity: high					anomaly_detection	5.94s	profile + IQR check → anomalies detected, severity: high				
root_cause_analysis	12.50s	FMSR → conf 0.621 → ALWAYS fires deep TSMF → conf 0.673 (+0.052) deep TSMF conf 0.621 → 0.673					root_cause_analysis	6.60s	FMSR → conf 0.621 $\geq \theta=0.60$ → gate skips deep TSMF gated ✓ conf 0.621				
validate_failure	<0.01s	policy check → work order needed					validate_failure	<0.01s	policy check → work order needed				
work_order_generation	0.39s	WO generated · tc = 1.0					work_order_generation	0.41s	WO generated · tc = 1.0				

← prev

play

next →

step

2 /

anomaly detection runs the same local profile check on both sides

executed deep TSMF fired inside step skipped / gated

Demo

Step-by-step replay of two SkillsAgent trajectories on the same fault-diagnosis scenario, showing the confidence gate in action

TSMF_430 · "We are building an early detection and remediation system for potential failures, such as condenser water side fouling, for Chiller 9... provide the guideline on work order generation?"

F — always deep TSMF			latency 20.1s	cost 26.5	tc 1.0	deep yes	E $\theta=0.60$ — confidence gate			latency 13.0s	cost 20.5	tc 1.0	deep no
● data_retrieval	<0.01s	fetch Chiller 9 sensor history (IoT CSV)					● data_retrieval	<0.01s	fetch Chiller 9 sensor history (IoT CSV)				
● anomaly_detection	7.22s	profile + IQR check → anomalies detected, severity: high					● anomaly_detection	5.94s	profile + IQR check → anomalies detected, severity: high				
● root_cause_analysis	12.50s	FMSR → conf 0.621 → ALWAYS fires deep TSMF → conf 0.673 (+0.052) deep TSMF conf 0.621 → 0.673					● root_cause_analysis	6.60s	FMSR → conf 0.621 $\geq \theta=0.60$ → gate skips deep TSMF gated ✓ conf 0.621 ✓				
● validate_failure	<0.01s	policy check → work order needed					● validate_failure	<0.01s	policy check → work order needed				
● work_order_generation	0.39s	WO generated · tc = 1.0					● work_order_generation	0.41s	WO generated · tc = 1.0				

← prev

play

next →

step

3 /

root cause analysis: FMSR diagnoses condenser fouling · pre-deep confidence = 0.621
F always fires deep TSMF here regardless · E checks: $0.621 \geq \theta=0.60 \rightarrow \text{skip}$

● executed ● deep TSMF fired inside step ● skipped / gated

Demo

Step-by-step replay of two SkillsAgent trajectories on the same fault-diagnosis scenario, showing the confidence gate in action

TFSM_430 · "We are building an early detection and remediation system for potential failures, such as condenser water side fouling, for Chiller 9... provide the guideline on work order generation?"

F — always deep TFSM			latency 20.1s	cost 26.5	tc 1.0	deep yes	E $\theta=0.60$ — confidence gate			latency 13.0s	cost 20.5	tc 1.0	deep no
● data_retrieval	<0.01s	fetch Chiller 9 sensor history (IoT CSV)					● data_retrieval	<0.01s	fetch Chiller 9 sensor history (IoT CSV)				
● anomaly_detection	7.22s	profile + IQR check → anomalies detected, severity: high					● anomaly_detection	5.94s	profile + IQR check → anomalies detected, severity: high				
● root_cause_analysis	12.50s	FMSR → conf 0.621 → ALWAYS fires deep TFSM → conf 0.673 (+0.052) deep TFSM conf 0.621 → 0.673					● root_cause_analysis	6.60s	FMSR → conf 0.621 $\geq \theta=0.60$ → gate skips deep TFSM gated ✓ conf 0.621 ✓				
● validate_failure	<0.01s	policy check → work order needed					● validate_failure	<0.01s	policy check → work order needed				
● work_order_generation	0.39s	WO generated · tc = 1.0					● work_order_generation	0.41s	WO generated · tc = 1.0				

← prev

play

next →

step

4 /

validate failure runs instantly on both sides — no TFSM involved

● executed ● deep TFSM fired inside step ● skipped / gated

Demo

Step-by-step replay of two SkillsAgent trajectories on the same fault-diagnosis scenario, showing the confidence gate in action

TSMF_430 · "We are building an early detection and remediation system for potential failures, such as condenser water side fouling, for Chiller 9... provide the guideline on work order generation?"

F — always deep TSMF			latency 20.1s	cost 26.5	tc 1.0	deep yes	E $\theta=0.60$ — confidence gate			latency 13.0s	cost 20.5	tc 1.0	deep no
data_retrieval	<0.01s	fetch Chiller 9 sensor history (IoT CSV)					data_retrieval	<0.01s	fetch Chiller 9 sensor history (IoT CSV)				
anomaly_detection	7.22s	profile + IQR check → anomalies detected, severity: high					anomaly_detection	5.94s	profile + IQR check → anomalies detected, severity: high				
root_cause_analysis	12.50s	FMSR → conf 0.621 → ALWAYS fires deep TSMF → conf 0.673 (+0.052) deep TSMF conf 0.621 → 0.673					root_cause_analysis	6.60s	FMSR → conf 0.621 $\geq \theta=0.60$ → gate skips deep TSMF gated ✓ conf 0.621 ✓				
validate_failure	<0.01s	policy check → work order needed					validate_failure	<0.01s	policy check → work order needed				
work_order_generation	0.39s	WO generated · tc = 1.0					work_order_generation	0.41s	WO generated · tc = 1.0				

← prev

play

next →

step 5 /

same work order generated · same tc = 1.0
F took 20.1s · E took 13.0s · saving: 7.1s (35%) · cost saving: 6.1 units (23%)

executed deep TSMF fired inside step skipped / gated

Demo

Step-by-step replay of two SkillsAgent trajectories on the same fault-diagnosis scenario, showing the confidence gate in action

TSMF_421 · "In May and June of 2020, numerous anomalies and alerts have been reported for Chiller 9 with equipment ID CWC04009. Could you review them and suggest appropriate work orders or actions?"

F — always deep TSMF			latency 19.9s	cost 26.5	tc 1.0	deep yes	D — no deep TSMF			latency 14.1s	cost 20.5	tc 1.0	deep no
data_retrieval	<0.01s	fetch Chiller 9 sensor history					data_retrieval	<0.01s	fetch Chiller 9 sensor history				
anomaly_detection	7.40s	anomalies detected, severity: high					anomaly_detection	7.14s	anomalies detected, severity: high				
root_cause_analysis	12.14s	FMSR → conf 0.573 → ALWAYS fires deep TSMF → conf 0.625 (+0.052) deep TSMF conf 0.573 → 0.625					root_cause_analysis	6.53s	FMSR → conf 0.573 · deep TSMF disabled (condition D) · stops at 0.573 conf 0.573				
validate_failure	<0.01s	policy check → work order needed					validate_failure	<0.01s	policy check → work order needed				
work_order_generation	0.38s	WO generated · tc = 1.0					work_order_generation	0.39s	WO generated · tc = 1.0				

← prev

play

next →

step

0 /

press play or click next to step through the trajectory

executed deep TSMF fired inside step skipped / gated

Demo

Step-by-step replay of two SkillsAgent trajectories on the same fault-diagnosis scenario, showing the confidence gate in action

TSMF_421 · "In May and June of 2020, numerous anomalies and alerts have been reported for Chiller 9 with equipment ID CWC04009. Could you review them and suggest appropriate work orders or actions?"

F — always deep TSMF			latency	19.9s	cost	26.5	tc	1.0	deep	yes	D — no deep TSMF			latency	14.1s	cost	20.5	tc	1.0	deep	no
● data_retrieval	<0.01s	fetch Chiller 9 sensor history									● data_retrieval	<0.01s	fetch Chiller 9 sensor history								
● anomaly_detection	7.40s	anomalies detected, severity: high									● anomaly_detection	7.14s	anomalies detected, severity: high								
● root_cause_analysis	12.14s	FMSR → conf 0.573 → ALWAYS fires deep TSMF → conf 0.625 (+0.052) deep TSMF conf 0.573 → 0.625									● root_cause_analysis	6.53s	FMSR → conf 0.573 · deep TSMF disabled (condition D) · stops at 0.573 conf 0.573								
● validate_failure	<0.01s	policy check → work order needed									● validate_failure	<0.01s	policy check → work order needed								
● work_order_generation	0.38s	WO generated · tc = 1.0									● work_order_generation	0.39s	WO generated · tc = 1.0								

← prev

play

next →

step

1 /

same scenario — anomaly review for Chiller 9 over May-June 2020

● executed ● deep TSMF fired inside step ● skipped / gated

Demo

Step-by-step replay of two SkillsAgent trajectories on the same fault-diagnosis scenario, showing the confidence gate in action

TSMF_421 · "In May and June of 2020, numerous anomalies and alerts have been reported for Chiller 9 with equipment ID CWC04009. Could you review them and suggest appropriate work orders or actions?"

F — always deep TSMF			latency	19.9s	cost	26.5	tc	1.0	deep	yes	D — no deep TSMF			latency	14.1s	cost	20.5	tc	1.0	deep	no
● data_retrieval	<0.01s	fetch Chiller 9 sensor history									● data_retrieval	<0.01s	fetch Chiller 9 sensor history								
● anomaly_detection	7.40s	anomalies detected, severity: high									● anomaly_detection	7.14s	anomalies detected, severity: high								
● root_cause_analysis	12.14s	FMSR → conf 0.573 → ALWAYS fires deep TSMF → conf 0.625 (+0.052) deep TSMF conf 0.573 → 0.625									● root_cause_analysis	6.53s	FMSR → conf 0.573 · deep TSMF disabled (condition D) · stops at 0.573 conf 0.573								
● validate_failure	<0.01s	policy check → work order needed									● validate_failure	<0.01s	policy check → work order needed								
● work_order_generation	0.38s	WO generated · tc = 1.0									● work_order_generation	0.39s	WO generated · tc = 1.0								

← prev

play

next →

step

3 /

F fires deep TSMF despite already having a valid diagnosis
D skips it entirely — no deep TSMF call at all · same diagnosis reached

● executed ● deep TSMF fired inside step ● skipped / gated

Demo

Step-by-step replay of two SkillsAgent trajectories on the same fault-diagnosis scenario, showing the confidence gate in action

TSMF_421 · "In May and June of 2020, numerous anomalies and alerts have been reported for Chiller 9 with equipment ID CWC04009. Could you review them and suggest appropriate work orders or actions?"

F — always deep TSMF			latency 19.9s	cost 26.5	tc 1.0	deep yes	D — no deep TSMF			latency 14.1s	cost 20.5	tc 1.0	deep no
data_retrieval	<0.01s	fetch Chiller 9 sensor history					data_retrieval	<0.01s	fetch Chiller 9 sensor history				
anomaly_detection	7.40s	anomalies detected, severity: high					anomaly_detection	7.14s	anomalies detected, severity: high				
root_cause_analysis	12.14s	FMSR → conf 0.573 → ALWAYS fires deep TSMF → conf 0.625 (+0.052) deep TSMF conf 0.573 → 0.625					root_cause_analysis	6.53s	FMSR → conf 0.573 · deep TSMF disabled (condition D) · stops at 0.573 conf 0.573				
validate_failure	<0.01s	policy check → work order needed					validate_failure	<0.01s	policy check → work order needed				
work_order_generation	0.38s	WO generated · tc = 1.0					work_order_generation	0.39s	WO generated · tc = 1.0				

← prev

play

next →

step

4 /

validate failure: both sides determine a work order is needed

● executed ● deep TSMF fired inside step ● skipped / gated

Demo

Step-by-step replay of two SkillsAgent trajectories on the same fault-diagnosis scenario, showing the confidence gate in action

TSMF_421 · "In May and June of 2020, numerous anomalies and alerts have been reported for Chiller 9 with equipment ID CWC04009. Could you review them and suggest appropriate work orders or actions?"

F — always deep TSMF			latency	19.9s	cost	26.5	tc	1.0	deep	yes	D — no deep TSMF			latency	14.1s	cost	20.5	tc	1.0	deep	no
data_retrieval	<0.01s	fetch Chiller 9 sensor history									data_retrieval	<0.01s	fetch Chiller 9 sensor history								
anomaly_detection	7.40s	anomalies detected, severity: high									anomaly_detection	7.14s	anomalies detected, severity: high								
root_cause_analysis	12.14s	FMSR → conf 0.573 → ALWAYS fires deep TSMF → conf 0.625 (+0.052) deep TSMF conf 0.573 → 0.625									root_cause_analysis	6.53s	FMSR → conf 0.573 · deep TSMF disabled (condition D) · stops at 0.573 conf 0.573								
validate_failure	<0.01s	policy check → work order needed									validate_failure	<0.01s	policy check → work order needed								
work_order_generation	0.38s	WO generated · tc = 1.0									work_order_generation	0.39s	WO generated · tc = 1.0								

← prev

play

next →

step 5 /

identical outcome: tc = 1.0, same work order
F: 19.9s · D: 14.1s · saving: 5.8s (29%) · deep TSMF added nothing here

executed deep TSMF fired inside step skipped / gated

Lessons Learned

Time budget: 1 minute. What surprised you? What didn't work? Be honest.

What worked

- Skill abstraction made the executor deterministic and trajectories diff-able across conditions.
- Confidence gating at $\theta=0.8$ hit ~50% Deep-TSFM skip rate — matching the redundancy audit's prediction.
- Reusing AssetOpsBench's native EvaluationAgent prompt avoided judge-prompt design overhead.

What didn't work

- Condition B as “ReAct”: implementation is a static unconditional pipeline. We relabeled it “static tool baseline” rather than rewriting.
- FMSR confidence proxy (heuristic until FMSR exposes native scores) is the largest source of measurement noise; future work: learned calibrator.
- Default $\theta=0.85$ in code drifted from the proposal's {0.7, 0.8, 0.9} sweep. We expanded the sweep to {0.5–0.95}.

Next Steps

Time budget: 30 seconds. Be specific — vague "more experiments" is not next steps.

- Replace the FMSR confidence heuristic with a learned calibrator trained on baseline trajectories — adaptive θ by asset type and failure mode.
- Extend the 54-scenario subset to all 141 scenarios and complete the LLM-judge cycle for every task category.
- Does the same skill/knowledge layer transfer to other AssetOpsBench domains (Pumps, Bearings, Engines) without per-asset re-tuning?
- We only focused on TSFM gating but would like to apply the same logic on the other agents - IoT, FMSR (Failure Mode Sensor Mapping), WO (Work Order)

Teamwork & Contributions

Time budget: 30 seconds. Who did what. Required slide — graded individually.

Member	Primary contributions	Tools / artifacts
Vera Mazeeva (vmm2146)	Skill abstraction layer, knowledge plugins, confidence evaluator; integration of AssetOpsBench MCP servers; assetopsbench scenario running → scoring pipeline	src/skills-knowledge-agent/ (code added from separate SkillsAgent repo that Mana and I worked on)
Shrey Arora (ska2153)	WatsonX EvaluationAgent judge, branch integration, ablation scoring pipeline	run_evaluation.py, score_ablation.py, team14-final integration branch
Sanskriti Shejwal (ss7561)	Full 150-scenario trajectory dump, TSFM redundancy auditor, auxiliary judge, Medium Article Draft	src/get_trajectories/, trajectories/, tsfm_report.{csv,json}
Mana Abbaszadeh (ma4845)	Wrapping skills with deepagents library, baseline experiments, results analysis	src/skills-knowledge-agent/*deep_agent, eval_results/ analysis
All	Ideation + proposal, report drafting, Medium Blog finishing	main.tex

Collaboration: weekly team syncs, shared GitHub fork (4 feature branches → team14-final integration branch), code review on every merge.

Thank you

Questions?

GitHub: github.com/shreyarora2198/AssetOpsBench (branch: team14-final)

Static results: `eval_results/`, `trajectories/`, `tsfm_report.{csv,json}`

Backup — Experimental Setup

Backup slide. Use for Q&A. Add more backup slides as needed.

scenarios		conditions		total runs	
54		12		648	
from tsfm_report		A, B, C, D, F, E×7		54 × 12 conditions	
cond.	label	planning	knowledge	deep TSFM	
baselines					
A	raw LLM — no tools	none	off	never	
B	static tool chain (ReAct stand-in)	none	off	never	
skill-augmented ablations					
C	planning only	LLM	off	never	
D	skills + knowledge, no deep TSFM	LLM	on	never	
F	skills + knowledge, always deep	LLM	on	always	
full system — θ sweep (7 conditions)					
E $\theta=0.50$	gate never fires (all conf > 0.5)	LLM	on	gated	
E $\theta=0.60$	selectively fires (16/54)	LLM	on	gated	
E $\theta=0.65$		LLM	on	gated	
E $\theta=0.70$		LLM	on	gated	
E $\theta=0.80$ ★	sweet spot — 50% reduction	LLM	on	gated	
E $\theta=0.90$		LLM	on	gated	
E $\theta=0.95$		LLM	on	gated	
hardware		scenario categories			
macOS Apple M-series · WatsonX cloud · Colab T4 GPU (granite-tsfm)		fault-diagnosis (32) · anomaly-detection (12) · forecasting (10)			

Backup — Theta Sweep Results

Backup slide. Use for Q&A. Add more backup slides as needed.

Theta sweep: TSFM reduction rate vs latency and task completion across conditions

baseline (AssetOpsBench) = 54 unnecessary TSFM calls · all E conditions run on the same 54 scenarios

condition	deep TSFM calls	reduction vs baseline	avg latency	task completion	fault-diag tc	anomaly tc
AssetOpsBench baseline	54 / 54	0%	—	—	—	—
A — raw LLM	0 / 54	100%	4.0s	0.191 ↓	0.104	0.305
B — tool baseline	0 / 54	100%	1.9s	0.308 ↓	0.291	0.333
D — skills, no deep TSFM	0 / 54	100%	10.8s	0.710 ↑	0.583	0.806
F — always deep TSFM	26 / 54	52%	13.6s	0.694 =	0.599	0.695
θ sweep — baseline (AssetOpsBench) had 54 unnecessary calls						
E θ=0.50 — gate never fires	0 / 54	100%	10.6s	0.688 ↓	0.599	0.667
E θ=0.60	16 / 54	70%	12.4s	0.713 ↑	0.599	0.778
E θ=0.65	26 / 54	52%	13.2s	0.691 =	0.552	0.806
E θ=0.70	29 / 54	46%	13.9s	0.713 ↑	0.568	0.861
E θ=0.80 ★	27 / 54	50%	13.5s	0.688 ↓	0.568	0.750
E θ=0.90	29 / 54	46%	14.0s	0.713 ↑	0.599	0.778
E θ=0.95	26 / 54	52%	13.2s	0.694 =	0.599	0.695

■ best trade-off (highlighted) reference conditions tc = task completion 0–1 · higher is better for both tc and reduction · lower is better for latency

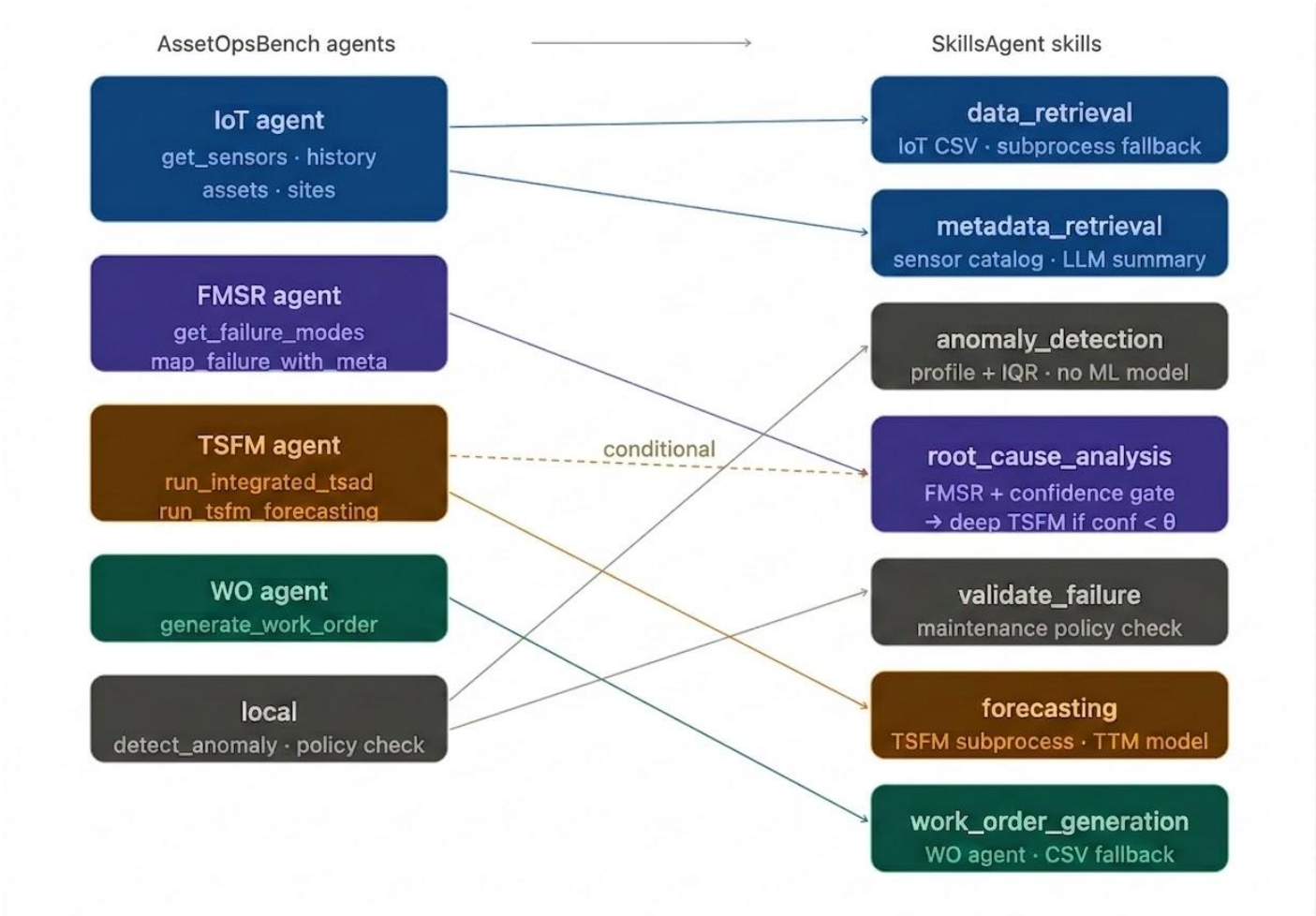
Backup — Calibrated Skills Cost (on Colab T4)

Backup slide. Use for Q&A. Add more backup slides as needed.

- **data_retrieval:** 0.4028
- **metadata_retrieval:** 3.6392
- **anomaly_detection:** 7.1758
- **root_cause_analysis:** 7.496
- **validate_failure:** 0.0
- **forecasting:** 4.9165
- **work_order_generation:** 0.7427
- **deep_tsfm:** 0.0004

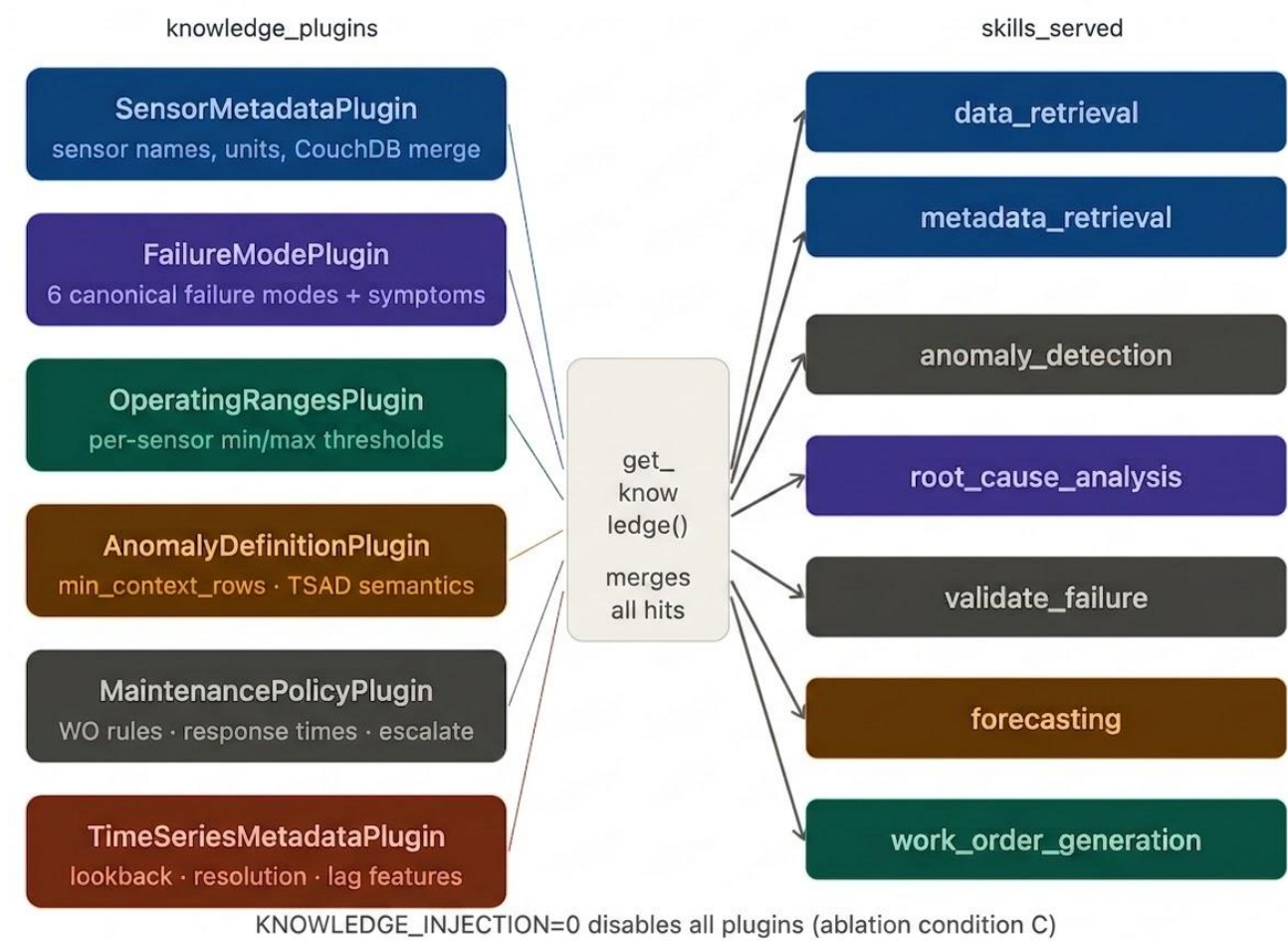
Backup — Skills Diagram

Backup slide. Use for Q&A. Add more backup slides as needed.



Backup — Knowledge Plugins Diagram

Backup slide. Use for Q&A. Add more backup slides as needed.



Backup — TSFM Workflow Diagram

Backup slide. Use for Q&A. Add more backup slides as needed.

